

Packet Tracer - Configure IPv6 ACLs

Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
PC1	NIC	2001:db8:1:10::10/64	fe80::1
PC2	NIC	2001:db8:1:11::11/64	fe80::1
Server3	NIC	2001:db8:1:30::30/64	fe80::30

Objectives

Part 1: Configure, Apply, and Verify an IPv6 ACL

Part 2: Configure, Apply, and Verify a Second IPv6 ACL

Instructions

Part 1: Configure, Apply, and Verify an IPv6 ACL

Logs indicate that a computer on the 2001:db8:1:11::0/64 network is repeatedly refreshing a web page. This is causing a Denial-of-Service (DoS) attack against **Server3**. Until the client can be identified and cleaned, you must block HTTP and HTTPS access to that network with an access list.

Step 1: Configure an ACL that will block HTTP and HTTPS access.

Configure an ACL named **BLOCK_HTTP** on **R1** with the following statements.

- Block HTTP and HTTPS traffic from reaching **Server3**.

```
R1(config)# deny tcp any host 2001:db8:1:30::30 eq www
R1(config)# deny tcp any host 2001:db8:1:30::30 eq 443
```

- Allow all other IPv6 traffic to pass.

Step 2: Apply the ACL to the correct interface.

Apply the ACL on the interface closest to the source of the traffic to be blocked.

Step 3: Verify the ACL implementation.

Verify that the ACL is operating as intended by conducting the following tests:

- = Open the **web browser** of **PC1** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should appear.
- = Open the **web browser** of **PC2** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should be blocked.
- = Ping from **PC2** to 2001:db8:1:30::30. The ping should be successful.

Part 2: Configure, Apply, and Verify a Second IPv6 ACL

The logs now indicate that your server is receiving pings from many different IPv6 addresses in a Distributed Denial of Service (DDoS) attack. You must filter ICMP ping requests to your server.

Step 1: Create an access list to block ICMP.

Configure an ACL named **BLOCK_ICMP** on **R3** with the following statements:

- Block all ICMP traffic from any hosts to any destination.
- Allow all other IPv6 traffic to pass.

Step 2: Apply the ACL to the correct interface.

In this case, ICMP traffic can come from any source. To ensure that ICMP traffic is blocked, regardless of its source or any changes that occur to the network topology, apply the ACL closest to the destination.

Step 3: Verify that the proper access list functions.

- Ping from **PC2** to 2001:db8:1:30::30. The ping should fail.
- Ping from **PC1** to 2001:db8:1:30::30. The ping should fail.
- Open the **web browser** of **PC1** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should display.

Cisco Packet Tracer - C:/Users/clcs96/Downloads/4.7.4-packet-tracer-configure-ipv6-acls.pka - ChristinaChalmers - 2026-04-21 15:47:07

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Activity Results

Congratulations ChristinaChalmers! You completed the activity.

Overall Feedback [Assessment Items](#) Connectivity Tests

Expand/Collapse All Show Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedback
Network				
R1				
ACLs		0	ACL	
✓ BLOCK_HTTP	Correct	40	IPv6 ACL Implementation	
Ports		0	Other	
GigabitEthernet0/1		0	Other	
✓ IPv6 Traffic Filter In	Correct	10	IPv6 ACL Implementation	
R3				
ACLs		0	ACL	
✓ BLOCK_ICMP	Correct	40	IPv6 ACL Implementation	
Ports		0	Other	
GigabitEthernet0/0		0	Other	
✓ IPv6 Traffic Filter Out	Correct	10	IPv6 ACL Implementation	

Score : 100/100
Item Count : 4/4

Component Items/Total Score
IPv6 ACL Implementation 4/4 100/100

Close

Fire weather stat... In effect

Search

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